Linear Regression

↳ Supervised ML algorithm

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Labelled data (having both i/p & o/p)

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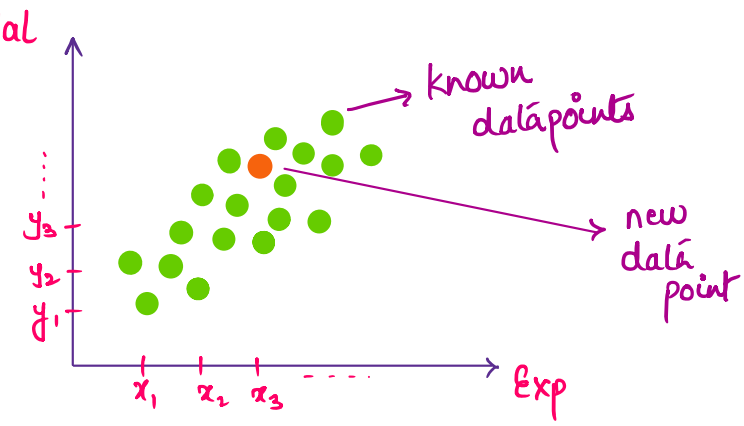
x y → continuous numeric values

Ex- Salary, Price, Age, etc.

For understanding linear regression,
let us take an example of salary prediction

Exp	Salary
x_1	y_1
x_2	y_2
⋮	⋮
x_n	y_n

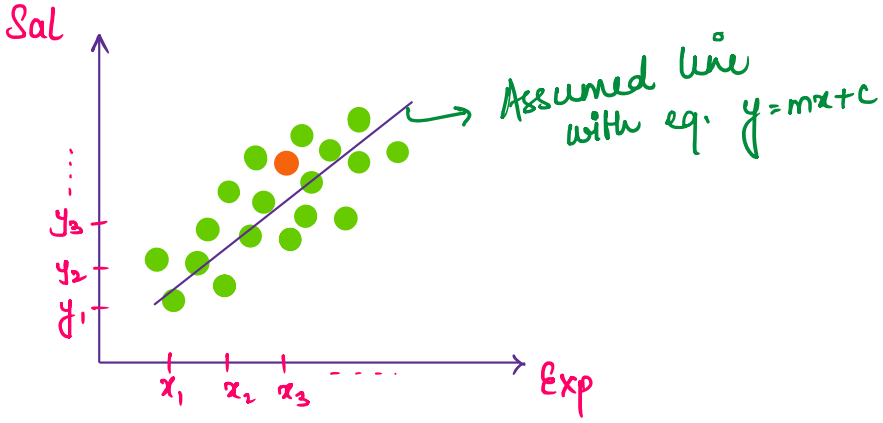
Plot the graph →



For predicting the value of y for a new x, we need an equation

Let us suppose an equation of a line for a 2D graph

$$y = mx + c$$



By drawing / assuming a line means that it is at minimum distance from all the data points. So, on drawing this line →

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"we assume an m & c value"

Let us suppose → $m = 0.1$ & $c = 1$

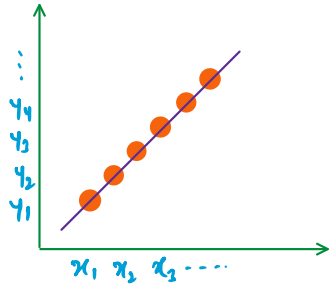
$$y = mx + c$$

$$y = 0.1x + 1$$

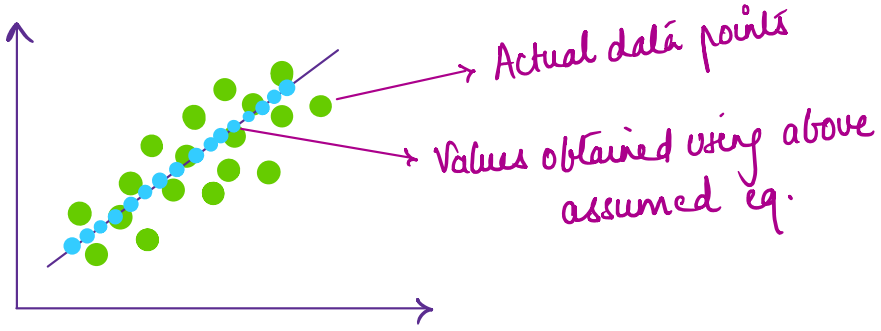
Using this eq. & substituting the value of $x \rightarrow x_1, x_2, x_3 \ldots$

we get multiple y's → $y_1, y_2, y_3 \ldots$

On drawing a graph →



Let us combine actual data & the above line graph



As we suppose that the assumed line (equation) is the one that helps in calculating the value of y for any x

So, for known x, it will calculate new values of y

then we can assume that these y will be the predicted y values.

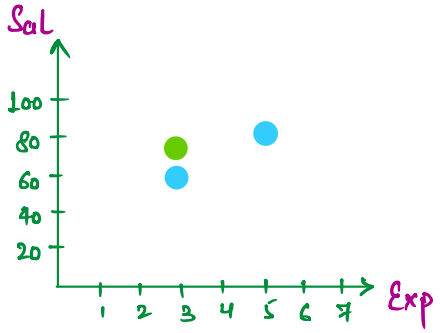
y values.

∴ We say that we draw a line among the data points so that it can help us in calculating predictions

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Let,

Exp	Sal
5	80
3	60
⋮	⋮



$\hat{y} = mx + c$

$m = 10, \quad c = 40$

$\hat{y} = 10 \times 3 + 40$

$\hat{y} = 70$

for $x = 3$, $y = 60$ (actual)

By using eq. for $x = 3$, $\hat{y} = 70$ (predicted)

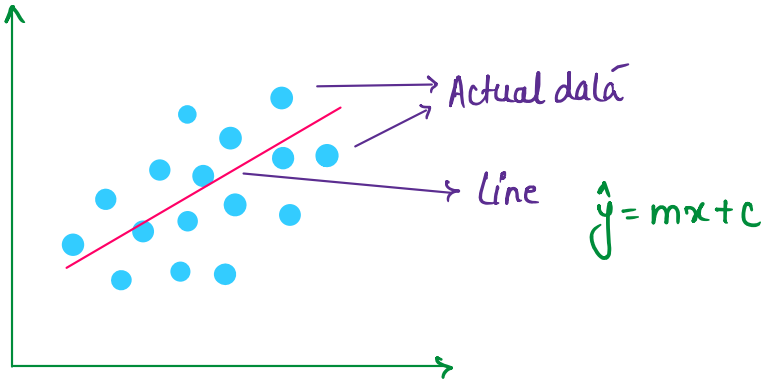
So, if dist. b/w y & $\hat{y}$ is small, it means prediction is good

By using above explanation, let's understand linear regression in detail →

Eq → $\hat{y} = mx + c$

m & c are unknown

[There can be any value of m & c but we need



of m & c but we need to find the best value of it]

Best slope & intercept → This is our goal

How to find it ?

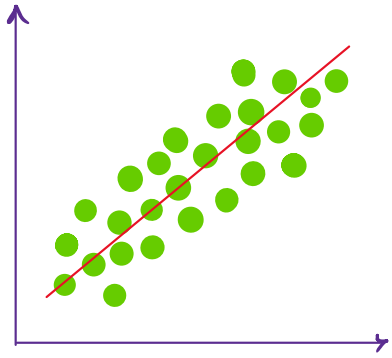
Steps :-

① Assume m & c

$$\hat{y} = mx + c$$

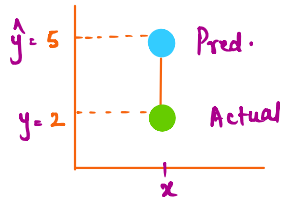
② Calculate $\hat{y}$ for each x

$$\hat{y}_1 = mx_1 + c \quad ; \quad \hat{y}_2 = mx_2 + c \quad ----$$



③ Calculate the distance b/w y & $\hat{y}$. We call this distance as error.

$$\text{Error} = y - \hat{y}$$



or

$$\varepsilon_1 = y_1 - \hat{y}_1 \;, \quad \varepsilon_2 = y_2 - \hat{y}_2 \;, \quad \varepsilon_3 = y_3 - \hat{y}_3 \quad ----$$

- We calculate errors for all data points
- The more the error, the bad is the prediction & vice versa

for a single error (overall error value)

$$\text{Mean Error (avg)} = \frac{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \text{-----} + \varepsilon_n}{n}$$

$$\text{Mean Error} = \frac{(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + (y_3 - \hat{y}_3) + \cdots}{n}$$

There is an issue with the above eq.

Let $y_1 = 8$ & $\hat{y}_1 = 1$ $\rightsquigarrow$ $y_1 - \hat{y}_1 = 8 - 1 = 7$

Also, $y_2 = 1$ & $\hat{y}_2 = 8$ $\rightsquigarrow$ $y_2 - \hat{y}_2 = 1 - 8 = -7$

$$\mathcal{E} = \frac{(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + \cdots}{n}$$

$$\mathcal{E} = \frac{7 - 7 + \cdots}{n}$$

There can be multiple such errors which can get cancelled

Mean Error ≈ 0 [Not necessary that the eq. of line is good]

In order to avoid this problem → convert -ve into +ve

- absolute (Modulus)
- square

a) Mean Absolute Error = $\frac{|\varepsilon_1| + |\varepsilon_2| + |\varepsilon_3| + \cdots}{n}$

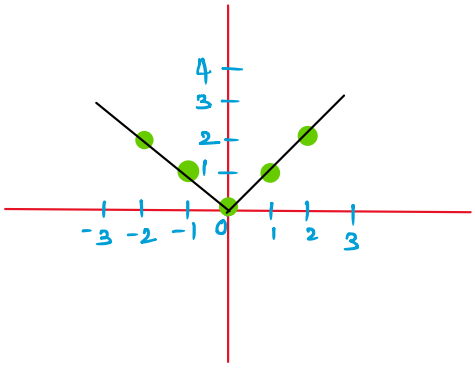
b) Mean Square Error = $\frac{\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \cdots}{n}$

Ques. which one to take?

Absolute

$y = |x|$

1. $x = 0$, $y = |0| = 0$
2. $x = 1$, $y = |1| = 1$
3. $x = -1$, $y = |-1| = 1$
4. $x = 2$, $y = |2| = 2$
5. $x = -2$, $y = |-2| = 2$

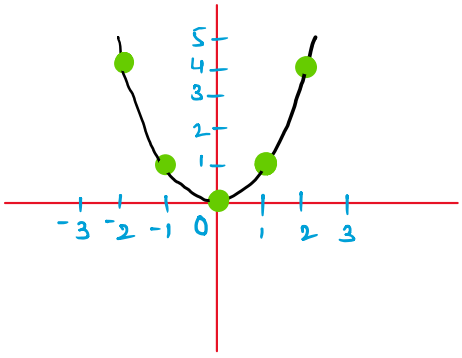


Absolute / Modulus function [discontinuous]

Square

$y = x^2$

1. $x = 0$, $y = 0^2 = 0$
2. $x = 1$, $y = 1^2 = 1$
3. $x = -1$, $y = (-1)^2 = 1$
4. $x = 2$, $y = 2^2 = 4$
5. $x = -2$, $y = (-2)^2 = 4$



Square function [Continuous]

We want a continuous function so we go for square values

∴ Mean Squared Error (MSE) $= \frac{\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \cdots + \varepsilon_n}{n}$

4. After calculating MSE, we will check if MSE is high or not

high or not

a) If MSE is high ⇝ eq. is bad

b) If MSE is low ⇝ eq. is good

↓

values of m & c is also good

↙

This good m & c is giving good predictions and the line drawn using that respective m & c is the best line.

Objective of Linear Regression → "To create a best fitting line"

* Challenge → Assuming m & c

[To get the perfect value of m & c]